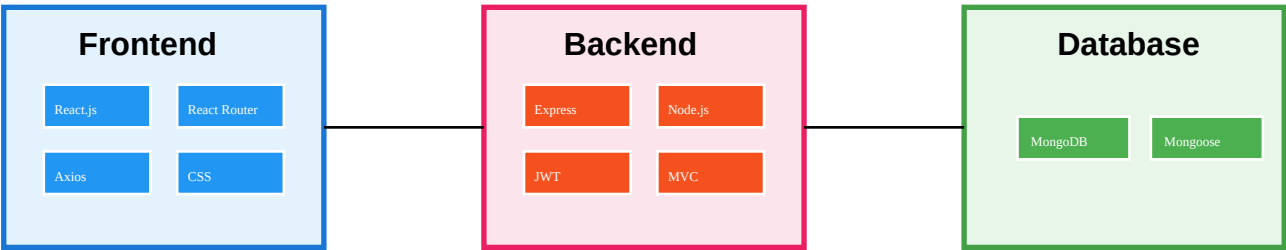


BOOK A DOCTOR - TECHNICAL ARCHITECTURE

The **Book a Doctor** application follows a modern three-tier MERN architecture. The frontend is developed using **React.js** with React Router, Axios and CSS to provide a responsive user interface. The backend is built using **Node.js** and **Express.js**, where REST APIs handle authentication, doctor management, patient management, profile management, and appointment booking. JWT authentication and bcrypt are used for secure login and password encryption. MongoDB with Mongoose provides scalable data storage for users and appointments. The project follows the **MVC architecture** with an additional Service Layer to separate business logic from controllers, making the application modular, secure, and maintainable.



MVC Architecture

Layer	Responsibility
Model	Defines MongoDB schemas using Mongoose (Users, Appointments).
View	React pages, components and dashboards for Admin, Doctor and Patient.
Controller	Receives API requests and returns responses.
Service	Implements business logic, CRUD operations, JWT and bcrypt.
Routes	Maps REST API endpoints to controllers.
Middleware	Performs authentication and role-based authorization.

Overall Request Flow

User → React UI → Axios → Express Routes → JWT Middleware → Controller → Service Layer → Mongoose → MongoDB → JSON Response → React UI